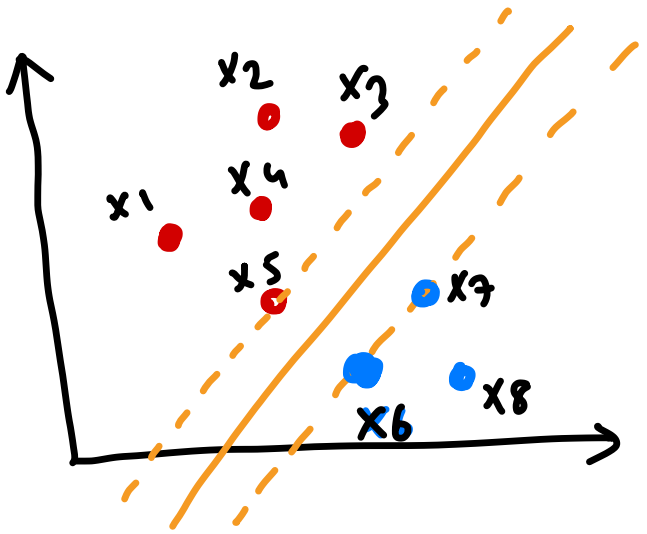
Let us assume we solve the dual problem $\Rightarrow \alpha^*$

$$w^* = \sum_{i=1}^{N} \alpha_i^* y_i x_i$$

most of α_i^*'s are zero

if $\alpha_i^* > 0$, x_i is called "support vector"

$$\alpha^* = \begin{bmatrix} \alpha_1^* \\ \alpha_2^* \\ \vdots \\ \alpha_N^* \end{bmatrix}$$



$\alpha_1^* = 0$ $\quad \alpha_5^* > 0$

$\alpha_2^* = 0$ $\quad \alpha_6^* > 0$

$\alpha_3^* = 0$ $\quad \alpha_7^* > 0$

$\alpha_4^* = 0$ $\quad \alpha_8^* = 0$

test data point

$$g(x) = w^{*T} \cdot x + w_0^* = \left[\left[\sum_{i=1}^{N} \alpha_i^* y_i x_i \right]^T \cdot x + w_0^* \right]$$

$$= \sum_{i=1}^{N} \alpha_i^* y_i x_i^T \cdot x + w_0^*$$

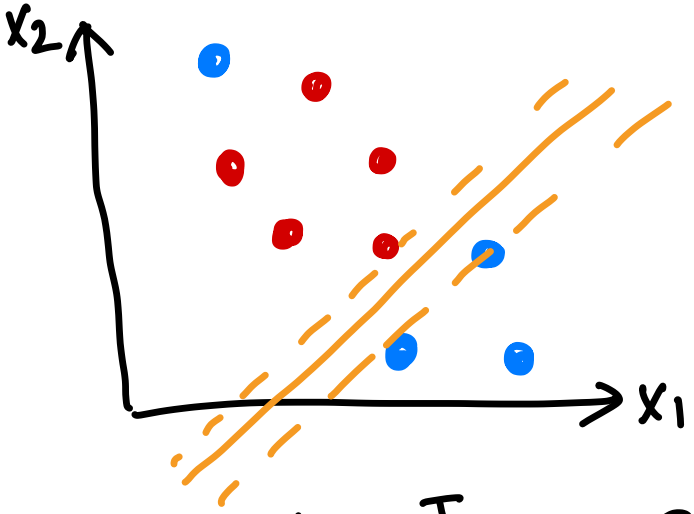
np.dot

only x_5, x_6 & x_7 will contribute

$k(x_i, x)$ ↳ kernel

We do not have to store x_1, x_2, x_3, x_4 & x_8 in the memory !!

Nonseparable Case



minimize $\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{N}\epsilon_i$

subject to: $\alpha_i\left[y_i\left[w^T.x_i + w_0\right] \geq 1-\epsilon_i\right] \forall_i$

$\beta_i[\ \epsilon_i \geq 0] \quad \forall_i$

$$L_p = \frac{1}{2}w^T.w + C\sum_{i=1}^{N}\epsilon_i - \sum_{i=1}^{N}\alpha_i\left[y_i\left[w^T.x_i+w_0\right]-1+\epsilon_i\right] - \sum_{i=1}^{N}\beta_i\epsilon_i$$

$$\frac{\partial L_p}{\partial w} = \frac{1}{2}.2.w - \sum_{i=1}^{N}\alpha_i y_i.x_i = 0 \Rightarrow \quad w = \sum_{i=1}^{N}\alpha_i y_i x_i$$

$$\frac{\partial L_p}{\partial w_0} = -\sum_{i=1}^{N}\alpha_i y_i = 0 \quad \Rightarrow \quad \sum_{i=1}^{N}\alpha_i y_i = 0$$

$$\frac{\partial L_p}{\partial \epsilon_i} = C - \alpha_i - \beta_i = 0 \quad \Rightarrow$$

$\alpha_i \geq 0$, $\beta_i \geq 0$

$$\alpha_i + \beta_i = C \quad \forall_i$$

$$\Downarrow$$

$$\boxed{0 \leq \alpha_i \leq C \quad \forall_i}$$

Exercise: Find the dual problem.

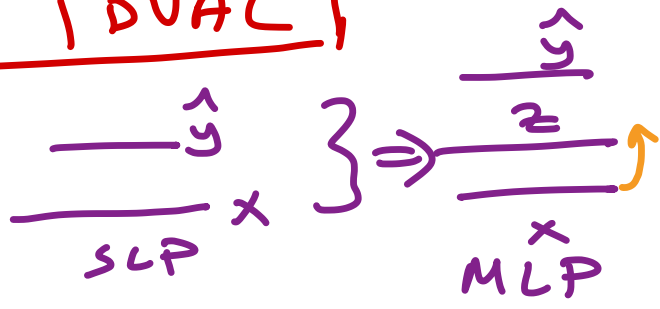
maximize $-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j y_i y_j x_i^T \cdot x_j + \sum_{i=1}^{N}\alpha_i$

subject to: $\sum_{i=1}^{N}\alpha_i y_i = 0$ (only difference compared to separable case)

$0 \leq \alpha_i \leq C \quad \forall i$

DUAL

$\sum_{i \in P}\alpha_i = \sum_{i \in N}\alpha_i$



Kernel Trick

$x \in \mathbb{R}^D \qquad z \in \mathbb{R}^Q$

usually $Q \gg D$

$\Phi : X \rightarrow Z$ (mapping function)

$D = 1$

$[x_i]$

$Q = 3$

$z_i = \Phi(x_i) = \begin{bmatrix} x_i \\ x_i^2 \\ x_i^3 \end{bmatrix}$

X domain

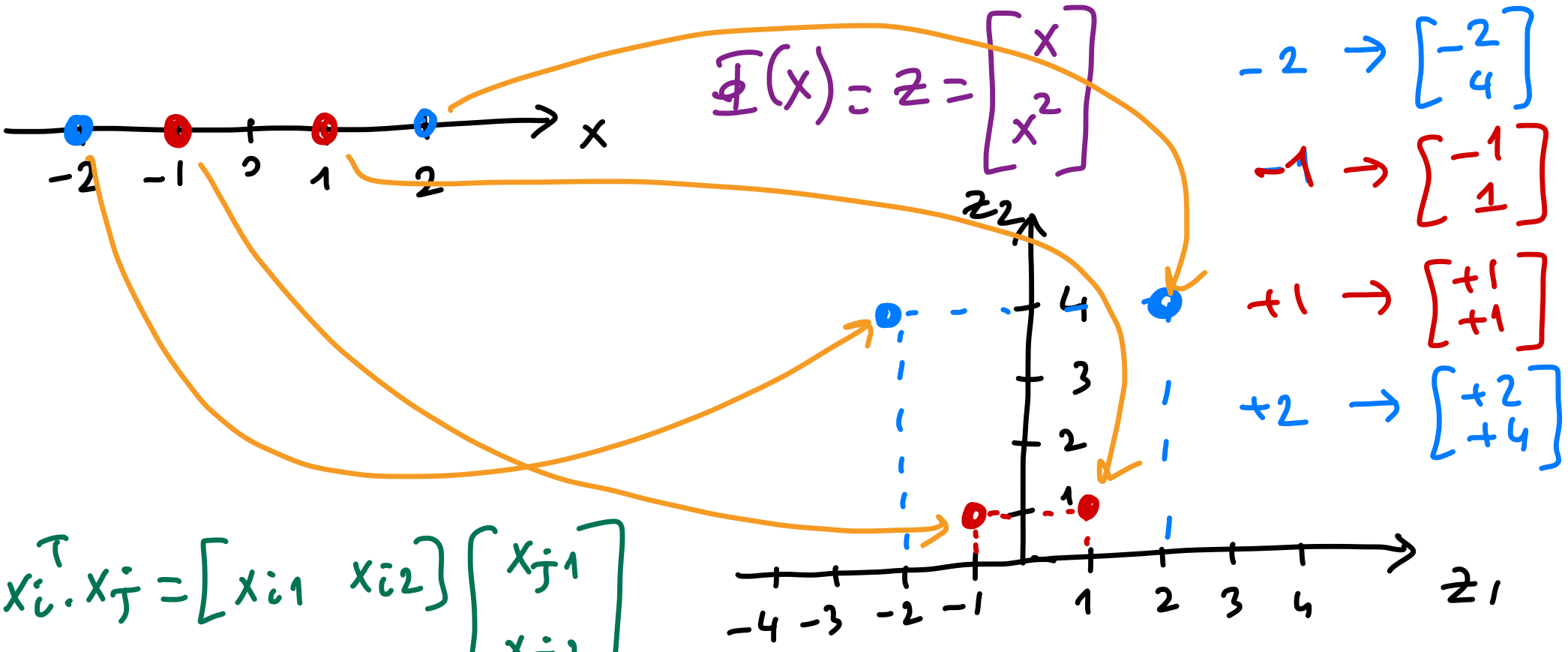
$w = \sum_{i=1}^{N}\alpha_i y_i x_i$

$g(x) = w^T \cdot x + w_0 = \sum_{i=1}^{N}\alpha_i y_i x_i^T \cdot x + w_0$

$w = \sum_{i=1}^{N}\alpha_i y_i z_i = \sum_{i=1}^{N}\alpha_i y_i \Phi(x_i)$

$g(x) = w^T \cdot z + w_0 = w^T \Phi(x) + w_0$

$= \sum_{i=1}^{N}\alpha_i y_i \boxed{\Phi(x_i)^T \Phi(x)} + w_0$



$$x_i^T \cdot x_j = \begin{bmatrix} x_{i1} & x_{i2} \end{bmatrix} \begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix}$$

$D = 2$

$$= x_{i1}x_{j1} + x_{i2}x_{j2}$$

$D=2 \qquad x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix}$

$Q=6 \qquad z_i = \begin{bmatrix} x_{i1}^2 \\ x_{i2}^2 \\ \sqrt{2}x_{i1}x_{i2} \\ \sqrt{2}x_{i1} \\ \sqrt{2}x_{i2} \\ 1 \end{bmatrix}$

$$x_i^T \cdot x_j = \begin{bmatrix} x_{i1} & x_{i2} \end{bmatrix} \begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix}$$

$$= x_{i1} \cdot x_{j1} + x_{i2} x_{j2}$$

$$\Phi(x_i)^T \cdot \Phi(x_j) = z_i^T \cdot z_j =$$

$$\begin{bmatrix} x_{i1}^2 & x_{i2}^2 & \sqrt{2}x_{i1}x_{i2} & \sqrt{2}x_{i1} & \sqrt{2}x_{i2} & 1 \end{bmatrix} \begin{bmatrix} x_{j1}^2 \\ x_{j2}^2 \\ \sqrt{2}x_{j1}x_{j2} \\ \sqrt{2}x_{j1} \\ \sqrt{2}x_{j2} \\ 1 \end{bmatrix}$$

$$(a+b+c)^2 = ((a+b)+c)^2$$

$$= (a+b)^2 + 2(a+b)c + c^2$$

$$= a^2 + 2ab + b^2 + 2ac + 2bc + c^2$$

$$(x_{i1}x_{j1} + x_{i2}x_{j2} + 1)^2$$

$$(x_i^T \cdot x_j + 1)^2$$

$$\Leftarrow = x_{i1}^2 x_{j1}^2 + x_{i2}^2 \cdot x_{j2}^2 + 2x_{i1}x_{i2}x_{j1}x_{j2} +$$

$$2x_{i1}x_{j1} + 2x_{i2}x_{j2} + 1$$